

Answers for Assignment 4

1 Problem 1 EM

Proof

The distribution and the log-likelihood of a mixture of Bernoulli's are given by:

$$p(x_i|\mu_k) = \prod_{j=1}^D \mu_{kj}^{x_{ij}} (1 - \mu_{kj})^{1-x_{ij}}$$

$$\log p(x_i|\mu_k) = \sum_{j=1}^D x_{ij} \log \mu_{kj} + (1 - x_{ij}) \log(1 - \mu_{kj})$$

Therefore, we get the following auxiliary function:

$$\begin{aligned} Q(\mu, \mu^{t-1}) &= E \left[\sum_{i=1}^N \log p(x_i, z_i|\mu^{t-1}) \right] \\ &= \sum_i \sum_k r_{ik} \log \pi_k + \sum_i \sum_k r_{ik} \log p(x_i|\mu_k) \\ &= \sum_i \sum_k r_{ik} \log \pi_k + \sum_i \sum_k r_{ik} \sum_j (x_{ij} \log \mu_{kj} + (1 - x_{ij}) \log(1 - \mu_{kj})) \end{aligned}$$

In M step, μ_{kj} is given by:

$$\mu_{kj} = \arg \max_{\mu_{kj}} Q(\mu, \mu^{t-1})$$

Let's take the derivative of $Q(\mu, \mu^{t-1})$ w.r.t. μ_{kj} .

$$\frac{\partial Q}{\partial \mu_{kj}} = \sum_i r_{ik} \left(\frac{x_{ij}}{\mu_{kj}} - \frac{1 - x_{ij}}{1 - \mu_{kj}} \right) \quad (1)$$

$$= \frac{1}{\mu_{kj}(1 - \mu_{kj})} \left(\sum_i r_{ik} x_{ij} - \mu_{kj} \sum_i r_{ik} \right) \quad (2)$$

Then, by derivative goes to 0.

$$\mu_{jk} = \frac{\sum_i r_{ik} x_{ij}}{\sum_i r_{ik}}$$

Beta prior is given by:

$$\begin{aligned}\text{Beta}(\alpha, \beta) &= \prod_k A \mu_{kj}^{\alpha-1} (1 - \mu_{kj})^{\beta-1} \quad \left(\because A = \frac{(\alpha + \beta - 1)!}{(\alpha - 1)!(\beta - 1)!} \right) \\ \log \text{Beta}(\alpha, \beta) &= \sum_k (\log A + (\alpha - 1) \log \mu_{kj} + (\beta - 1) \log(1 - \mu_{kj}))\end{aligned}$$

In M step of MAP estimation, μ_{kj} is given by:

$$\mu_{kj} = \arg \max_{\mu_{kj}} Q' := Q(\mu, \mu^{t-1}) + \log \text{Beta}(\alpha, \beta, \mu_{kj}) \quad (3)$$

The derivative of $\log \text{Beta}(\alpha, \beta, \mu_{kj})$ w.r.t. μ_{kj} is:

$$\frac{\partial}{\partial \mu_{kj}} \log \text{Beta}(\alpha, \beta, \mu_{kj}) = \frac{\alpha - 1}{\mu_{kj}} - \frac{\beta - 1}{1 - \mu_{kj}} \quad (4)$$

$$= \frac{\alpha - 1 - \mu_{kj}(\alpha + \beta - 2)}{\mu_{kj}(1 - \mu_{kj})} \quad (5)$$

By combining equations 2,3,5 and $\frac{\partial Q'}{\partial \mu_{kj}} = 0$, we get the following equation:

$$\mu_{kj} = \frac{(\sum_i r_{ik} x_{ij}) + \alpha - 1}{(\sum_i r_{ik}) + \alpha + \beta - 2} \quad (6)$$

2 Problem 2 AUC

Proof:

Obviously, we just need to prove $\sum_{i=1}^{m^+} \sum_{j=1}^{m^-} u(e_{ij}) = \sum_{i=1}^{m^+} \text{rank}_i - (m^+)(m^+ + 1)/2$.

If $m^- = 0$, then $LHS = \sum_{i=1}^{m^+} i - \frac{(m^+)(m^++1)}{2} = 0 \Leftrightarrow \sum_{i=1}^{m^+} i = \frac{(m^+)(m^++1)}{2}$. Assume that $g(\mathbf{x}_i^+)$ is increasing with i . Choose one $g(\mathbf{x}_1^-)$ and assume that $g(\mathbf{x}_{k-1}^+) < g(\mathbf{x}_1^-) < g(\mathbf{x}_k^+)$, then $LHS = RHS$ because:

$$\begin{aligned}LHS &= \sum_{i=1}^{m^+} u(e_{i1}) = \sum_{i=1}^{m^+} u(g(\mathbf{x}_i^+) - g(\mathbf{x}_1^-)) = m^+ - k + 1 \\ RHS &= \sum_{i=1}^{m^++1} i - \frac{(m^+)(m^+ + 1)}{2} = \frac{(m^+ + 1)(m^+ + 2)}{2} - k - \frac{(m^+)(m^+ + 1)}{2} = m^+ - k + 1\end{aligned}$$

If $LHS = RHS$ holds for all $g(\mathbf{x}_i^+)$ and $g(\mathbf{x}_j^-)$ when $j = n$, then if $j = n + 1$,

$$RHS_{n+1} - LHS_{n+1} = \left(\sum_{i=1}^{m^+} rank_i\right)_{n+1} - \left(\sum_{i=1}^{m^+} rank_i\right)_n - \sum_{i=1}^{m^+} u(g(\mathbf{x}_i) - g(\mathbf{x}_{n+1}^-))$$

Since $\exists k'$ such that $g(\mathbf{x}_{k'-1}^+) < g(\mathbf{x}_{n+1}^-) < g(\mathbf{x}_{k'}^+)$, $(rank_i)_{n+1} - (rank_i)_n = 1$ for $i \geq k$. Besides, $g(\mathbf{x}_i^+) > g(\mathbf{x}_{n+1}^-)$ for $i \geq k$, $\therefore RHS_{n+1} - LHS_{n+1} = (n - k' + 1) - (n - k' + 1) = 0$.

By induction, we can conclude that $LHS = RHS, \forall n \in \{1, \dots, m^-\}$, so $\sum_{i=1}^{m^+} \sum_{j=1}^{m^-} u(e_{ij}) = \sum_{i=1}^{m^+} rank_i - (m^+)(m^+ + 1)/2$.

3 Problem 3 PCA

Solution: Here is an example, which is not unique.

$$\mathbf{u} = (-0.4, 0.8, 0.2, 0.2, -1.3), \mathbf{Y}^{(n)} = \frac{1}{10}(\mathbf{X}^{(n)} - \mathbf{u})$$

$$\mathbf{Y}^T \mathbf{Y} = \begin{bmatrix} 0.304 & 0.082 & -0.002 & -0.082 & -0.012 \\ 0.082 & 0.376 & -0.216 & 0.104 & 0.104 \\ -0.002 & -0.216 & 0.356 & 0.076 & -0.084 \\ -0.082 & 0.104 & 0.076 & 0.556 & 0.116 \\ -0.012 & 0.104 & -0.084 & 0.116 & 0.221 \end{bmatrix}$$

$$\text{The orthogonal basis } (u_1, \dots, u_5) = \begin{bmatrix} -0.312 & 0.169 & 0.878 & 0.298 & -0.026 \\ 0.606 & -0.333 & 0.176 & 0.395 & 0.579 \\ 0.620 & 0.083 & 0.405 & -0.580 & -0.329 \\ -0.331 & -0.157 & 0.144 & -0.646 & 0.654 \\ 0.167 & 0.910 & -0.111 & 0.030 & 0.359 \end{bmatrix} \text{ are the}$$

unit-length principle components of X .

The eigenvalues are 0.082, 0.153, 0.307, 0.593, 0.677.

Choose the last two largest eigenvalues $\mathbf{U} = (\mathbf{u}_4, \mathbf{u}_5)$ for PCA. With the equation

$$\mathbf{z} = \mathbf{U}^T(\mathbf{x} - \mathbf{u})\tilde{\mathbf{X}} = \mathbf{u} + \mathbf{U}\mathbf{z},$$

the projections are:

$$\begin{pmatrix} 1.9818 \\ -3.1319 \end{pmatrix}, \begin{pmatrix} 4.4965 \\ -0.6075 \end{pmatrix}, \begin{pmatrix} -1.4035 \\ 1.9732 \end{pmatrix}, \begin{pmatrix} -2.8786 \\ 0.4956 \end{pmatrix}, \begin{pmatrix} 0.4823 \\ -0.7201 \end{pmatrix}, \\ \begin{pmatrix} 1.7424 \\ 1.8758 \end{pmatrix}, \begin{pmatrix} 0.5395 \\ 5.3831 \end{pmatrix}, \begin{pmatrix} -4.4578 \\ -0.5917 \end{pmatrix}, \begin{pmatrix} -1.0718 \\ -4.4691 \end{pmatrix}, \begin{pmatrix} 0.5691 \\ -0.2074 \end{pmatrix}$$